

ARC ROS WorkShop 1

This guide covers:

- Installing Docker on Ubuntu, Windows, and macOS
- Basic Docker commands
- Pulling and running a ROS 2 Humble + TurtleBot3 VNC Docker image
- Pushing a Docker image
- Basic Ubuntu commands
- Basic ROS 2 commands
- Running the TurtleBot3 Gazebo world simulation
- Moving TurtleBot3 using `ros2 topic pub`

1. Install Docker

1.1 Ubuntu

Recommended method: install Docker Engine from Docker's official `apt` repository.

Step 1: Remove old/conflicting packages

```
sudo apt remove docker.io docker-compose docker-compose-v2 docker-doc podman-docker containerd runc
```

It is okay if some of these packages are not installed.

Step 2: Install required packages

```
sudo apt update
sudo apt install -y ca-certificates curl
```

Step 3: Add Docker's official GPG key

```
sudo install -m 0755 -d /etc/apt/keyrings

sudo curl -fsSL https://download.docker.com/linux/ubuntu/gpg \
-o /etc/apt/keyrings/docker.asc

sudo chmod a+r /etc/apt/keyrings/docker.asc
```

Step 4: Add Docker's repository

```
echo \
"deb [arch=$(dpkg --print-architecture) signed-by=/etc/apt/keyrings/docker.asc] \
https://download.docker.com/linux/ubuntu \
${. /etc/os-release && echo "${UBUNTU_CODENAME:-$VERSION_CODENAME}"} stable" \
| sudo tee /etc/apt/sources.list.d/docker.list > /dev/null
```

Step 5: Install Docker Engine

```
sudo apt update

sudo apt install -y docker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-compose-plugin
```

Step 6: Test Docker

```
sudo docker run hello-world
```

If Docker is installed correctly, you should see a message saying that the Docker installation is working.

Optional: Run Docker without `sudo`

```
sudo usermod -aG docker $USER
```

Then log out and log back in, or reboot.

Test again:

```
docker run hello-world
```

1.2 Windows

Recommended method: install Docker Desktop.

Requirements

- Windows 10 or Windows 11
- WSL 2 enabled
- Hardware virtualization enabled in BIOS/UEFI
- Administrator permissions during installation

Steps

1. Download Docker Desktop for Windows from Docker's official website.
2. Run the installer.
3. Keep **Use WSL 2 instead of Hyper-V** selected, unless you specifically need Hyper-V.
4. Restart your computer if requested.
5. Open Docker Desktop.
6. Wait until Docker says it is running.
7. Open PowerShell or Windows Terminal and test:

```
docker --version
docker run hello-world
```

Useful Windows check

Check installed WSL distributions:

```
wsl --list --verbose
```

Update WSL:

```
wsl --update
```

1.3 macOS

Recommended method: install Docker Desktop.

Steps

1. Download Docker Desktop for Mac from Docker's official website.
2. Choose the correct installer:
 - Apple Silicon: M1, M2, M3, M4
 - Intel chip
3. Open the `.dmg` file.
4. Drag Docker into the `Applications` folder.
5. Open Docker from `Applications`.
6. Allow required permissions.
7. Open Terminal and test:

```
docker --version
docker run hello-world
```

2. Simple Docker Cheat Sheet

Check Docker version

```
docker --version
```

Show Docker system information

```
docker info
```

Download an image

```
docker pull IMAGE_NAME:TAG
```

Example:

```
docker pull ubuntu:22.04
```

List downloaded images

```
docker images
```

List running containers

```
docker ps
```

List all containers

```
docker ps -a
```

Stop a container

```
docker stop CONTAINER_NAME_OR_ID
```

Remove a container

```
docker rm CONTAINER_NAME_OR_ID
```

Remove an image

```
docker rmi IMAGE_NAME:TAG
```

Execute a command inside a running container

```
docker exec -it CONTAINER_NAME bash
```

Show container logs

```
docker logs CONTAINER_NAME_OR_ID
```

Build an image from a Dockerfile

```
docker build -t IMAGE_NAME:TAG .
```

Example:

```
docker build -t my-ros-image:latest .
```

Login to Docker Hub

```
docker login
```

3. Pull the ROS 2 Humble + TurtleBot3 VNC Image

Use this command to pull the image:

```
docker pull pjmrbotics/ros2-turtlebot3-vnc:latest
```

If you want a specific tag, replace `latest` with the tag name:

```
docker pull pjmrbotics/ros2-turtlebot3-vnc:tagname
```

4. Run the ROS 2 TurtleBot3 VNC Image

Run the container:

```
docker run --rm -it \
  --name ros2-turtlebot3-vnc \
  --shm-size=2g \
  -p 6080:80 \
  -p 5900:5900 \
  pjmrbotics/ros2-turtlebot3-vnc:latest
```

What these options mean

Option	Meaning
<code>--rm</code>	Remove the container after it exits
<code>-it</code>	Run interactively with a terminal
<code>--name ros2-turtlebot3-vnc</code>	Give the container a readable name
<code>--shm-size=2g</code>	Increase shared memory, useful for GUI/simulation workloads
<code>-p 6080:80</code>	Map browser-based VNC/noVNC to host port <code>6080</code>
<code>-p 5900:5900</code>	Map standard VNC to host port <code>5900</code>
<code>pjmrbotics/ros2-turtlebot3-vnc:latest</code>	Docker image to run

Open the GUI

After the container starts, open this in your browser:

```
http://localhost:6080
```

If you are using a VNC client, connect to:

```
localhost:5900
```

6. Very Simple Ubuntu Commands

Show current directory

```
pwd
```

List files

```
ls
```

List files with details

```
ls -la
```

Change directory

```
cd DIRECTORY_NAME
```

Go to home directory

```
cd ~
```

Go up one directory

```
cd ..
```

Create a directory

```
mkdir my_folder
```

Remove an empty directory

```
rmdir my_folder
```

Remove a file

```
rm file.txt
```

Remove a directory and everything inside it

```
rm -r my_folder
```

Be careful with `rm -r`; deleted files may not be recoverable.

Copy a file

```
cp source.txt destination.txt
```

Move or rename a file

```
mv old_name.txt new_name.txt
```

Create an empty file

```
touch notes.txt
```

View a file

```
cat file.txt
```

Edit a file with nano

```
nano file.txt
```

Save in nano:

```
Ctrl + O  
Enter  
Ctrl + X
```

Clear terminal

```
clear
```

Check running processes

```
ps aux
```

Kill a process by PID

```
kill PID_NUMBER
```

Update package list

```
sudo apt update
```

Install a package

```
sudo apt install PACKAGE_NAME
```

Example:

```
sudo apt install git
```

Check IP address

```
ip a
```

Download a file

```
wget URL
```

or:

```
curl -O URL
```

7. Very Simple ROS 2 Commands

Before using ROS 2, source the ROS 2 environment:

```
source /opt/ros/humble/setup.bash
```

If using a workspace:

```
source ~/turtlebot3_ws/install/setup.bash
```

Check ROS 2 command help

```
ros2 --help
```

List running ROS 2 nodes

```
ros2 node list
```

Get information about a node

```
ros2 node info /NODE_NAME
```

Example:

```
ros2 node info /robot_state_publisher
```

List topics

```
ros2 topic list
```

List topics with message types

```
ros2 topic list -t
```

Echo a topic

```
ros2 topic echo /TOPIC_NAME
```

Example:

```
ros2 topic echo /cmd_vel
```

Show topic information

```
ros2 topic info /TOPIC_NAME
```

Example:

```
ros2 topic info /scan
```

Show message type definition

```
ros2 interface show geometry_msgs/msg/Twist
```

List services

```
ros2 service list
```

List actions

```
ros2 action list
```

Show ROS 2 package executables

```
ros2 pkg executables PACKAGE_NAME
```

Example:

```
ros2 pkg executables turtlebot3_teleop
```

Run a ROS 2 executable

```
ros2 run PACKAGE_NAME EXECUTABLE_NAME
```

Example:

```
ros2 run turtlebot3_teleop teleop_keyboard
```

Launch a ROS 2 launch file

```
ros2 launch PACKAGE_NAME LAUNCH_FILE.py
```

Example:

```
ros2 launch turtlebot3_gazebo turtlebot3_world.launch.py
```

8. Run the TurtleBot3 World Simulation

Inside the Docker container, open a terminal and run:

```
source /opt/ros/humble/setup.bash
```

If the TurtleBot3 workspace exists:

```
source ~/turtlebot3_ws/install/setup.bash
```

Set the TurtleBot3 model:

```
export TURTLEBOT3_MODEL=burger
```

Other valid model names are usually:

```
export TURTLEBOT3_MODEL=waffle
```

or:

```
export TURTLEBOT3_MODEL=waffle_pi
```

Launch the TurtleBot3 world:

```
ros2 launch turtlebot3_gazebo turtlebot3_world.launch.py
```

If everything is working, Gazebo should open with the TurtleBot3 world.

9. Move the Robot with the Keyboard

Open a second terminal inside the container:

```
docker exec -it ros2-turtlebot3-vnc bash
```

Then run:

```
source /opt/ros/humble/setup.bash
source ~/turtlebot3_ws/install/setup.bash
export TURTLEBOT3_MODEL=burger
ros2 run turtlebot3_teleop teleop_keyboard
```

Use the keyboard keys shown in the terminal to move the robot.

10. Move the Robot with `ros2 topic pub`

TurtleBot3 listens for velocity commands on:

```
/cmd_vel
```

The message type is:

```
geometry_msgs/msg/Twist
```

A `Twist` message has:

- `linear.x`: forward/backward speed
- `angular.z`: turning speed

Move forward

```
ros2 topic pub /cmd_vel geometry_msgs/msg/Twist \
"{linear: {x: 0.2, y: 0.0, z: 0.0}, angular: {x: 0.0, y: 0.0, z: 0.0}}"
```

Move backward

```
ros2 topic pub /cmd_vel geometry_msgs/msg/Twist \
"{linear: {x: -0.2, y: 0.0, z: 0.0}, angular: {x: 0.0, y: 0.0, z: 0.0}}"
```

Turn left

```
ros2 topic pub /cmd_vel geometry_msgs/msg/Twist \
"{linear: {x: 0.0, y: 0.0, z: 0.0}, angular: {x: 0.0, y: 0.0, z: 0.5}}"
```

Turn right

```
ros2 topic pub /cmd_vel geometry_msgs/msg/Twist \
"{linear: {x: 0.0, y: 0.0, z: 0.0}, angular: {x: 0.0, y: 0.0, z: -0.5}}"
```

Stop the robot

```
ros2 topic pub /cmd_vel geometry_msgs/msg/Twist \
"{linear: {x: 0.0, y: 0.0, z: 0.0}, angular: {x: 0.0, y: 0.0, z: 0.0}}"
```

11. Publish Velocity Continuously

The commands above may publish repeatedly depending on your ROS 2 CLI behavior. To publish at a fixed rate, use `--rate`.

Move forward continuously at 10 Hz

```
ros2 topic pub --rate 10 /cmd_vel geometry_msgs/msg/Twist \
"{linear: {x: 0.2, y: 0.0, z: 0.0}, angular: {x: 0.0, y: 0.0, z: 0.0}}"
```

Stop the command with:

```
Ctrl + C
```

Then publish a stop command:

```
ros2 topic pub --once /cmd_vel geometry_msgs/msg/Twist \
"{linear: {x: 0.0, y: 0.0, z: 0.0}, angular: {x: 0.0, y: 0.0, z: 0.0}}"
```

12. Common Troubleshooting

Docker command says permission denied on Ubuntu

Run:

```
sudo usermod -aG docker $USER
```

Then log out and log back in.

Container name already exists

Remove the old container:

```
docker rm ros2-turtlebot3-vnc
```

Or use another name:

```
docker run --name another-container-name ...
```

Port already in use

Check what is using the port:

```
sudo lsof -i :6080
```

Use different host ports:

```
docker run --rm -it \
  --name ros2-turtlebot3-vnc \
  --shm-size=2g \
  -p 6081:80 \
  -p 5901:5900 \
  pjmrobotics/ros2-turtlebot3-vnc:latest
```

Then open:

```
http://localhost:6081
```

ROS 2 command not found

Source ROS 2:

```
source /opt/ros/humble/setup.bash
```

TurtleBot3 package not found

Source the workspace:

```
source ~/turtlebot3_ws/install/setup.bash
```

If the package is not installed in the image, install or build the TurtleBot3 simulation packages.

Gazebo is slow or crashes

Try:

```
docker run --rm -it \
  --name ros2-turtlebot3-vnc \
  --shm-size=4g \
  -p 6080:80 \
  -p 5900:5900 \
  pjmrobotics/ros2-turtlebot3-vnc:latest
```

13. Quick Full Workflow

Terminal 1: Start container

```
docker run --rm -it \
  --name ros2-turtlebot3-vnc \
  --shm-size=2g \
  -p 6080:80 \
  -p 5900:5900 \
  pjmrobotics/ros2-turtlebot3-vnc:latest
```

Open:

```
http://localhost:6080
```

Terminal 2: Launch TurtleBot3 world

```
docker exec -it ros2-turtlebot3-vnc bash
source /opt/ros/humble/setup.bash
source ~/turtlebot3_ws/install/setup.bash
export TURTLEBOT3_MODEL=burger
ros2 launch turtlebot3_gazebo turtlebot3_world.launch.py
```

Terminal 3: Move robot

```
docker exec -it ros2-turtlebot3-vnc bash
source /opt/ros/humble/setup.bash
source ~/turtlebot3_ws/install/setup.bash
export TURTLEBOT3_MODEL=burger
ros2 topic pub --rate 10 /cmd_vel geometry_msgs/msg/Twist \
  "{linear: {x: 0.2, y: 0.0, z: 0.0}, angular: {x: 0.0, y: 0.0, z: 0.0}}"
```

Stop with:

```
Ctrl + C
```

Then stop the robot:

```
ros2 topic pub --once /cmd_vel geometry_msgs/msg/Twist \
  "{linear: {x: 0.0, y: 0.0, z: 0.0}, angular: {x: 0.0, y: 0.0, z: 0.0}}"
```

14. References

- Docker Engine on Ubuntu: <https://docs.docker.com/engine/install/ubuntu/>
- Docker Desktop on Windows: <https://docs.docker.com/desktop/setup/install/windows-install/>
- Docker Desktop on Mac: <https://docs.docker.com/desktop/setup/install/mac-install/>
- Docker image pull reference: <https://docs.docker.com/reference/cli/docker/image/pull/>
- Docker image push reference: <https://docs.docker.com/reference/cli/docker/image/push/>
- ROS 2 Humble topics tutorial: <https://docs.ros.org/en/humble/Tutorials/Beginner-CLI-Tools/Understanding-ROS2-Topics/Understanding-ROS2-Topics.html>
- TurtleBot3 simulation manual: <https://emanual.robotis.com/docs/en/platform/turtlebot3/simulation/>
- TurtleBot3 model environment variable: https://emanual.robotis.com/docs/en/platform/turtlebot3/export_turtlebot3_model/